OIL PUMP MONITOR

Real-Time Vibration Analysis & Anomaly Detection

DATABRICKS SOLUTION ACCELERATOR

Powered by Databricks Apps • Unity Catalog • Serverless

REAL-TIME VIBRATION MONITORING & AI-POWERED FAULT DETECTION FOR FRACKING PUMPS

Oil Pump Monitor delivers live vibration analysis, FFT spectrum visualization, anomaly detection, and an AI operations assistant for upstream oil & gas fracking operations in the North Dakota Bakken Formation.

◆ BUSINESS VALUE

- **Prevent catastrophic pump failures** with real-time vibration amplitude and frequency monitoring
- **Identify fault signatures early** — bearing faults, cavitation, imbalance, and overspeed detected automatically
- **Reduce unplanned downtime** with 2-second refresh live metrics across 6 Bakken field pumps
- **Enable data-driven maintenance** with FFT spectrum analysis revealing harmonic patterns
- **Accelerate fault diagnosis** with Foundation Model API-powered AI operations assistant

◆ CHALLENGES ADDRESSED

- **Undetected vibration anomalies** — bearing faults and cavitation causing costly unplanned shutdowns
- **Manual spectrum analysis** — technicians interpreting FFT plots without automated pattern recognition
- **No real-time visibility** — periodic manual readings missing time-critical vibration excursions
- **Disconnected alerting** — sensor alarms not correlated with fault type or recommended action
- **Limited field coverage** — monitoring only at the wellhead with no centralized fleet view

◆ WHO IT'S FOR

🔧 **Maintenance Technician**
Live vibration waveforms, FFT spectrum analysis, and anomaly alert feed

📊 **Reliability Engineer**
Fault signature library, trend analysis, and predictive failure patterns

🗺️ **Field Supervisor**
Geographic pump map, fleet-wide status, and severity-based alert prioritization

🤖 **Operations Manager**
AI-powered diagnostics, root cause analysis, and corrective action recommendations

6	4	2s	50 Hz
Pumps Monitored	Fault Types	Refresh Rate	FFT Range

◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the Oil Pump Monitor repo to your local machine
2. **Build Frontend**
   Run cd frontend && npm install && npm run build
3. **Deploy the App**
   Run databricks apps deploy to launch the monitor
4. **Monitor Pumps**
   Auto-seeds demo data and starts vibration simulator

◆ GETTING THE MOST OUT OF OIL PUMP MONITOR

📡 **Watch Live Metrics**
Monitor real-time vibration amplitude, RPM, temperature, and pressure for all 6 pumps

📊 **Analyze FFT Spectrum**
Examine frequency-domain patterns to identify bearing faults, cavitation, and imbalance

🗺️ **View Field Map**
See geographic distribution of pump locations across the Bakken field

🤖 **Ask Genie AI**
Chat with the operations assistant for fault diagnosis and corrective actions

◆ TARGET INDUSTRY

Upstream Oil & Gas — Fracking pump operations in the Bakken Formation, North Dakota. Applicable to any reciprocating or centrifugal pump fleet requiring vibration-based condition monitoring.

⚠ **DEMO ONLY** — This application is a demonstration and solution accelerator. Please refer to the included LICENSE file for full terms of use.

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